Step 4: Find Q_3

89,138,164,215,296

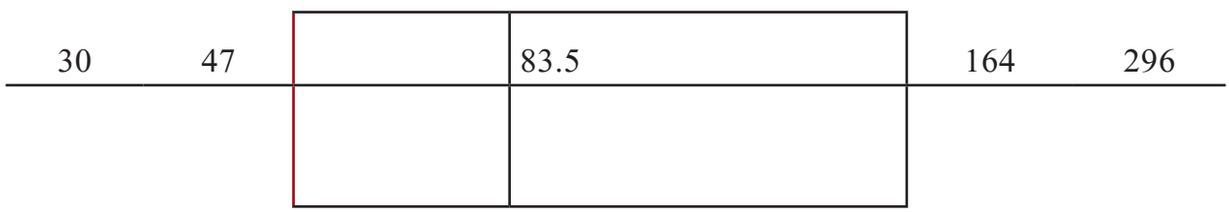
$Q_3 = 164$

Step 5: Find the minimum and maximum values.

Step 6: Locate the lowest value, Q_1, median, Q_3 and the highest value on the scale.

Step 7: Draw a box through Q_1 and Q_3

Box plot



Example 6.21

Construct a box –whisker plot for the following data

96, 151, 167, 185, 200, 220, 246, 269, 238, 252, 297, 105, 123, 178, 202

Solution:

Step 1: Arrange the data in code

96,105,123,151,167,178,185,200,202,220,238,246,252,269,297.

Step 2: Find the Median

8th term Median = 200

Step 3: Find Q_1 (middle of previous terms of 200)

96,105,123,151,167,178,185

$Q_1 = 151$

Step 4 : Find Q_3 (middle of successive terms of 200)

202,220,238,246,252,269,297

$Q_3 = 246$

Step 5: Minimum value = 96, Maximum Value = 297

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